

Important Terms and Ideas for Describing Artificial Intelligence

There are several terms experts use to describe computer systems in the field of artificial intelligence.

Recently, the French News Agency (AFP) defined some of the common terms and ideas used in that field.

Here is a version for English learners:

Artificial intelligence

The first term is “artificial intelligence.”

When asked what artificial intelligence is, the AI-powered ChatGPT system says that the term means “the simulation of human intelligence in machines that are programmed to think, learn and make decisions”.

AI's main quality or characteristic is taking in large amounts of data and then processing it using methods from statistics.

AI involves using ideas from many fields including computing, mathematics, languages, psychology, and others.

Currently, the technology is being used heavily for investigating health issues, translating human languages, and predicting problems in machine tools and self-driving cars. But AI is affecting many fields of business and industry.

FILE - OpenAI CEO Sam Altman attends an event to pitch AI for businesses in Tokyo, Japan February 3, 2025. REUTERS/Kim Kyung-Hoon

Algorithm

A second important term is “algorithm.”

An algorithm is important to all computer operations. It is a series of steps or instructions followed by a computer program to get a result.

Algorithms can give rules for an AI's behavior, helping it to realize the objectives of computer program developers.

Unlike a simple computer program, AI algorithms permit a computer system to “learn” for itself.

Machine learning

A third important term is “machine learning.”

Machine learning is one method that researchers have used in their efforts to produce artificial intelligence.

Machine learning lets computers learn from data without being directly programmed on what results to produce.

In recent years, the field of neural networks has given important results.

In a neural network, connections between some nodes are strengthened and others weakened as the system learns and makes changes.

Learning can be "supervised." This means the system learns to put new data into specific groups based on a model. For example, the system could learn to identify spam in an email or other messaging programs.

FILE - Visitors watch a projection at the Kunstpalast art museum in Duesseldorf, Germany, Tuesday, May 2, 2023. Turkish artist Refik Anadol used a machine learning framework to let Artificial Intelligence create new generative landscapes. (AP Photo/Martin Meissner)

"Unsupervised" learning permits the system to independently discover new areas or ways of doing things. These discoveries in the available data might not have been immediately clear.

An example would be letting an online store identify buying trends in sales data.

"Reinforcement" learning adds a process of repeated trial-and-error. In this process, the system is rewarded based on its outcomes, causing it to learn and improve.

One example might be a self-driving vehicle whose objective is to reach its destination as quickly as possible but also safely. That requirement would lead it to learn to stop at red lights although it requires additional time.

Deep learning

Deep learning owes its name to its use of many layers of neural networks.

Raw data is examined by each layer in turn at growing levels of abstraction.

Geoffrey Hinton received the 2024 Nobel Prize in Physics. Hinton is credited with developing deep learning. Hinton received the prize along with 1980s neural-network developer John Hopfield.

Francis Bach, head of France's SIERRA statistical learning laboratory, said this about deep learning: "The more layers you have, the more complex behavior can become, and the more complex the behavior can be, the easier it is to learn a desired behavior efficiently."

The method might help lead to scientific discoveries.

FILE - Text from the ChatGPT page of the OpenAI website is shown in this photo, in New York, Feb. 2, 2023. Anthropic, ChatGPT-maker OpenAI and other developers of large language models say they're hard at work to make them more truthful. (AP Photo/Richard Drew, File)

Language models

We now turn to large language models (LLMs).

These might be the most popular example of generative AI. Large language models power tools like OpenAI's ChatGPT or Google's Gemini.

Such systems are able to write long papers, answer legal questions or even produce a cake recipe based on their statistical models.

But the technology is still new. LLM's can suffer from "hallucinations"- the creation of content that is false or incorrect.

Artificial general intelligence

A final important term is artificial general intelligence (AGI) - one the big goals of the whole AI field.

AGI suggests the unrealized dream of a machine able to reproduce all human processes of human thinking.

People who push the idea include OpenAI chief Sam Altman and his competitors at Anthropic. They consider such a system to be within reach.

The goal is to use large amounts of data and processing power to train LLMs that are increasingly powerful.

But critics say that LLM technology has important limits, including its ability to reason.

Maxime Amblard, computing professor at France's University of Lorraine, told AFP last year, "LLMs do not work like human beings."

Amblard added that humans, as flesh-and-blood -intelligent beings, are "sense-making machines" with different abilities from today's computer systems.

I'm Anna Matteo. And I'm John Russell.

Vocabulary

simulation - n. การจำลอง การแสดงการทำงานของระบบหรือกระบวนการหนึ่งโดยใช้อีกระบบหนึ่ง

statistics - n. พหุพจน์. สถิติ สาขาวิชาการประมวลผลข้อมูลเชิงตัวเลขเพื่ออธิบายกระบวนการและสิ่งต่างๆ

neural - adj. เกี่ยวกับเส้นประสาท ที่เกี่ยวข้องกับสมองหรือเส้นประสาท

node - n. โหนด จุดที่ส่วนเล็กๆ เริ่มต้นหรือรวมศูนย์

spam - n. สแปม ข้อความที่ไม่พึงประสงค์ (เช่น อีเมล) ที่มักมีวัตถุประสงค์ทางการค้า

trend - n. แนวโน้ม ทิศทางของการเคลื่อนไหวหรือการเปลี่ยนแปลง

abstraction - n. การสรุปแนวคิด การสร้างแนวคิด

efficiently - adv. อย่างมีประสิทธิภาพ อย่างสำเร็จลุล่วง มีความสามารถ หรือมีผลลัพธ์ที่เหมาะสม

Reading Comprehension Quiz

1. According to paragraphs 2 to 4, what is the primary objective of this article?

- A. To introduce the latest developments in AI technology.
- B. To provide a version of common AI terms for English learners.
- C. To critique the definitions provided by the French News Agency.
- D. To explain how AI systems are built and operated.

2. Based on the definition provided by the AI-powered ChatGPT system, what is the fundamental nature of artificial intelligence?

- A. The development of physical robots that mimic human actions.
- B. The simulation of human intelligence within machines programmed for cognitive tasks.
- C. A branch of mathematics focused on data processing and statistics.
- D. The creation of self-improving computer algorithms.

3. What is highlighted as the primary characteristic or quality of AI operations in the article?

- A. Its ability to generate creative content instantly.
- B. Its reliance on human input for every decision.
- C. Its capacity to take in and process large amounts of data using statistics.
- D. Its primary use in translating human languages.

4. Which of the following academic fields is NOT explicitly mentioned as contributing ideas to Artificial Intelligence?

- A. Computing
- B. Mathematics
- C. Literature
- D. Psychology

5. How do AI algorithms primarily distinguish themselves from simple computer programs, according to the text?

- A. They are designed by larger teams of developers.
- B. They require more extensive computer hardware.
- C. They allow a computer system to "learn" independently.
- D. They only operate on specific types of data.

6. What is the defining characteristic of "machine learning" as described in the article?

- A. It requires direct programming for every specific result.
- B. It enables computers to learn from data without explicit programming for results.
- C. It is solely focused on strengthening neural network connections.
- D. It involves human experts supervising all data processing.

7. In "supervised" learning, what is the main goal for the system when presented with new data?

- A. To generate entirely new and original content.
- B. To discover unknown patterns without prior guidance.
- C. To put new data into specific groups based on an existing model.
- D. To continuously improve its internal algorithms through trial-and-error.

8. What is a key capability of "unsupervised" learning?

- A. It trains the system using a reward system for desired outcomes.
- B. It allows the system to independently identify new areas or ways of doing things in data.
- C. It requires human supervisors to label all input data.
- D. It specializes in translating human languages.

9. What process is central to "reinforcement" learning, according to the article?

- A. Manual input of new data by human operators.
- B. Direct programming of desired outcomes.
- C. A process of repeated trial-and-error with rewards based on outcomes.

D. Strengthening and weakening connections within a neural network.

10. Deep learning earns its name due to which particular characteristic?

- A. Its ability to solve very profound philosophical problems.
- B. Its use of many layers of neural networks.
- C. Its capacity for understanding human emotions deeply.
- D. Its origin in ancient, deeply hidden algorithms.

11. According to Francis Bach, head of France's SIERRA statistical learning laboratory, what is the benefit of increasing the number of layers in deep learning?

- A. It reduces the computational power required.
- B. It makes the behavior more complex, leading to more efficient learning.
- C. It simplifies the underlying mathematical models.
- D. It limits the system to only specific types of data processing.

12. What is a noted drawback or limitation of large language models (LLMs) mentioned in the text?

- A. Their inability to translate human languages accurately.
- B. Their tendency to generate "hallucinations" or false content.
- C. Their excessive cost, making them inaccessible for most users.
- D. Their requirement for constant human supervision.

13. What is described as the "unrealized dream" associated with Artificial General Intelligence (AGI)?

- A. A machine that can perfectly predict future events.
- B. A machine capable of reproducing all human processes of thinking.
- C. The development of AI that can manage global economies.
- D. An AI system that can operate without any human intervention.

14. Critics point out a significant limit of LLM technology in the context of achieving AGI. What is it?

- A. Its limited memory capacity.
- B. Its inability to process visual information.
- C. Its limited ability to reason.
- D. Its slow processing speed.

15. How does Professor Maxime Amblard describe human beings when contrasting them with current computer systems?

- A. As purely logical decision-makers.
- B. As "sense-making machines" with distinct abilities.
- C. As less efficient at data processing.
- D. As primarily driven by statistical models.